



Fig. 2: A fast hopped signal captured by a real-time spectrum analyser. The left side displays the user-defined frequency mask trigger, while the right spectrogram displays the captured hopping signal

system components all must work in concert to ensure optimal performance. Because software actively alters SDR operating parameters, there are countless hardware/software combinations that can cause errors. Modulation and filtering transients, distortion, non-linear power effects, pulse aberrations, frequency tuning and settling, power supply coupling, digital to RF couplings, and software-dependent phase errors are also common.

Designing fast frequency synthesisers presents a significant challenge as well. For example, a joint tactical information distribution system (JTIDS) deployed by the US armed forces working in an L-Band TDMA network operates at 38,461.5 hops per second. This means the frequency synthesiser has to hop from one frequency to the next, settle and communicate in less than 26µs time; system transient responses must be settled in just a few hundred nanoseconds to enable error-free communication.

Impaired modulation quality due to frequency settling of hopped carriers is one of the primary sources for poor transmitter quality and low system data rates. In the past, designers were able to use conventional test equipment to demodulate stationary carriers located at the centre frequency of their modulation analyser. However, conventional test equipment are not capable of demodulating today's wide-

band hopped signals. Because these signals hop over the band of operation, analyses of off-centre frequencies are required to ensure optimal modulation quality.

The dynamic generation of RF waveforms through DSP and the integration of digital and RF circuits—often on the same integrated circuit (IC)—also create issues not seen in traditional RF transceiver designs. These include modulation transients, non-linear effects of amplifiers, and digital-to-RF crosstalk, to name a few.

The performance of SDR transmitters must be verified with measurements that are beyond the traditional RF transmitter conformance tests. Simply passing these tests does not ensure a device will work properly, and system behaviour must be carefully and thoroughly observed since software is continually changing system parameters. Truly addressing these challenges requires SDR designers to fully analyse and characterise their systems.

Discovery of true system behaviour is important to identify potential RF spectrum anomalies. As system parameters change over time, performing frequency-selective triggering is necessary to pinpoint the instant a transient event occurs. Performing time-correlated analysis in multiple domains is required to determine the specific cause of each problem. Capturing the entire event seamlessly into memory is valuable for subsequent analysis, as it can be difficult to recreate the conditions under which the transient occurred. These advanced troubleshooting methods of verifying signal performance over time, combined with traditional conformance tests performed under steady-state conditions, are necessary for comprehensive SDR testing.

## Verifying performance and troubleshooting at the system level

Developing a verified system architecture design is vital to the success of a modern communication system. The more the access points that are tested and verified, the less likely the issues that will manifest during the last system integration phase. Also, the later the issues are addressed in the development and integration cycle, the more expensive they will be to resolve. Some of the major contributors to system failures are DSP, RF circuitry and the controlling software. Verification debug tools greatly aid system designers in effectively discovering problems.

Once an error has been identified, it must be isolated and understood. To isolate a problem and determine its root cause, it is important to time-correlate the error back through the signal path. Since the signal information changes form in an SDR design—from digital bits to continuously variable analogue voltages—several pieces of test equipment may be needed to diagnose the exact source of problems. Because the problem may occur at any point in the signal path, and memory capacity in oscilloscopes and logic analysers is limited, the ability to simultaneously trigger multiple test instruments and capture the exact moment in time that the event occurs is important. This requires that each instrument be able to trigger in its domain (logic analysers for digital triggers, oscilloscopes for time-domain amplitude triggers and spectrum analysers for frequency-domain triggers).

An integrated, end-to-end test system comprising real-time spectrum analyser (RTSA), arbitrary waveform generator, oscilloscope and logic analyser can be invaluable for testing SDRs. Select instruments from leading test and measurement vendors are able to work in unison—with cross-triggering and time-correlated subsystem views—to verify SDR per-